Sailab Banik

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SUMMARY

Software Engineer with 3 years of production experience at ZF Group, the last 2 delivering LLM-powered document intelligence: a platform adopted by **20+ business teams** that cut manual review by **60%**, and automated extraction across **3,000+ scanned engineering drawings**. Specializes in **Python, FastAPI, React, RAG pipelines, and LLM validation**, owning delivery from requirements to deployment.

WORK EXPERIENCE

Software Engineer II Oct 2024 – Present
ZF Group *Chennai, India*

- Built the AI layer of **PrimeAI**, an internal document processing platform covering extraction, user-defined models, and retrieval; adopted by **20+ business teams**, it reduced manual review by **60%**, saving **2,000+ hours per quarter**.
- Designed hybrid model routing between **locally served Ollama models** and frontier APIs, keeping restricted documents inside internal infrastructure while meeting accuracy targets through retrieval rather than larger models.
- Delivered a **PLM drawing extraction** solution end to end: gathered requirements directly with the client, independently researched DIN and ISO drawing standards, and built the pipeline on **Azure OpenAI**, automating dimension capture across **3,000+ scanned drawings**.
- Led the frontend and owned product flows from UI to API and database; designed **role- and group-based access control** from the ground up, enabling rollout to departments with strict data-access requirements.
- Rebuilt the team's CI/CD process and introduced **React Testing Library and Vitest** suites, shortening release cycles by **40%**, lowering production hotfixes by **35%**, and reducing manual testing by **60%**.

Software Engineer (Graduate Trainee) Aug 2023 – Sep 2024
ZF Group *Chennai, India*

- Resolved **10+ high-priority production bugs** on a business-critical legacy platform, cutting user-reported issues by **45%**.
- Led the pre-audit **security hardening** of internal applications, remediating injection and input-validation findings; **every application passed review**.
- Reworked operations scheduling workflows, improving throughput by **30%**.

PROJECTS

ContextDesk Lab: Context Engineering Playground for Redis Iris *Python, FastAPI, Redis Iris, Next.js, TypeScript, Docker*

- Reduced repeat-question latency from **4.65s to 917ms with 0 tokens billed** by serving semantically equivalent prompts from **LangCache**; cache misses still report their similarity score.
- Built an AI support console that returns an execution record with every response, orchestrating **Redis Iris LangCache, Agent Memory, and Context Retriever**, covered by **50 offline tests**.

FocusTrack: Work Tracker with AI Reviews | **focustrack.sailab.dev** *Next.js, TypeScript, Supabase, PostgreSQL, Vercel*

- Built and shipped a tracker that classifies work sessions as execution or preparation and writes daily, weekly, and monthly **AI reports** from those numbers; a database claim caps AI calls at **1 per user per period**.
- Secured **Supabase** Postgres and Auth with **row-level security** on every user table, verified by **60 pgTAP tests**; set up DNS and a GitHub Actions pipeline running **150+ unit tests** before each **Vercel** deploy.

QueueLens: Distributed Job Queue with Live Dashboard *Python, FastAPI, Redis, PostgreSQL, Next.js, Docker*

- Built a producer–consumer job queue: a **FastAPI** producer persists jobs in **PostgreSQL** and hands job IDs to background workers over a **Redis** list (LPUSH/BRPOP), with a **Next.js** dashboard tracing every job through its lifecycle live.
- Hardened it with **retries and exponential backoff**, Redis-backed **per-client rate limiting**, and a queue cap returning 429/503 with Retry-After; committing state before enqueueing eliminated API–worker race conditions.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript (ES6+), SQL, HTML, CSS
AI/ML: Generative AI, RAG Pipelines, Large Language Models (LLMs), Vision-Language Models (VLMs), Structured Outputs, LLM Evaluation, Semantic Caching, Context Engineering, Azure OpenAI, Ollama, LangChain, Redis Iris (LangCache, Agent Memory)
Backend: FastAPI, REST APIs, Pydantic, SQLAlchemy, PostgreSQL, Supabase (Auth, RLS), Redis, OAuth2 / SSO (Microsoft Entra ID, MSAL)
Frontend: React.js, Next.js, TypeScript, Redux Toolkit, Tailwind CSS, Material UI, React Testing Library, Vitest, MSW
DevOps & Tools: Docker, Docker Compose, CI/CD Pipelines, GitHub Actions, Azure DevOps, Vercel, DNS, Git

EDUCATION

ICFAI University Tripura Aug 2019 – Jul 2023
*B.Tech in Computer Science and Engineering – **CGPA: 9.64*** *Agartala, India*

AWARDS & CERTIFICATIONS

Budding Star Award — ZF Group, Recognition for Business Impact in the first year *2024*
System Design Cohort — ChaiCode: distributed systems, load balancing, caching, sharding, queues Credential | *Jun 2026*